

Chapter

The Endocrine System

1. Priya tried to match the hormones with their effect on the human body. She tabulated the pairs as follows: SQP 2025

Hormone	Effect on the body
P	Stimulates conversion of Glycogen to Glucose
Q	Reabsorption of water by the Nephrons.

Identify the correct pair of hormones.

- (a) P — Insulin, Q — Glucagon.
- (b) P — Adrenaline, Q — Oxytocin.
- (c) P — Glucagon, Q — Vasopressin.
- (d) P — Vasopressin, Q — Adrenaline.

2. The hormone that affects urination is: MAIN 2024

- (a) Adrenaline
- (b) Vasopressin
- (c) Oestrogen
- (d) Thyroxine

3. Gigantism and Acromegaly are due to : SEM I, 2021

- (a) Hyposecretion of Thyroxine.
- (b) Hyposecretion of Growth hormone
- (c) Hypersecretion of Thyroxine

(d) Hypersecretion of Growth hormone

4. Gigantism and Acromegaly are due to : SEM II, 2021

(a) Hypersecretion of Growth hormone

(b) Hypersecretion of Thyroxine

(c) Hyposecretion of Growth hormone

(d) Hyposecretion of Thyroxine

5. Name the endocrine gland that secretes Thyroxine. Give any one function of Thyroxine. SEM II, 2021

6. Give the full form of the abbreviation ACTH. Which gland secretes this hormone ? SEM II, 2021

7. Name the form in which glucose is stored in liver. COMP 2023

8. The chemical messengers. MAIN 2021

9. Thyroid gland secretes adrenaline. SQP 2021

10. A patient with diabetes visits a doctor, who explains that a specific part of the body contains clusters of cells that regulate blood sugar levels by producing hormones like insulin. Islets of Langerhans are located in :

(a) Liver

(b) Pituitary gland

(c) Spleen

(d) Pancreas SEM II, 2021

11. The child frightened by the loud noise, runs quickly to his mother. Which substance helps in this response? MAIN 2023

(a) Insulin

(b) Adrenaline

(c) Amylase

(d) Protease

12. Maintains glucose level in the blood MAIN 2022

13. Answer the following briefly : SQP 2021

(i) If you stand to make your maiden speech before a large audience, your mouth dries up and heart beat increases. What brings about these changes?

(ii) What are endocrine glands ?

(iii) Explain briefly – 'Glucose is normally not found in urine'.

(iv) Why are hormonal reactions generally slower than nervous reactions.

(v) A diabetic person has high concentration of sugar in urine-what is this condition known as ?

14. Expand the abbreviation — ADH. Give one example of a diuretic substance.
SQP 2025

15. Why is Adrenaline called 'The Emergency Hormone'? SQP 2025

16. What are Tropic hormones? Give one example . SQP 2025

17. Name the mineral elements required for MAIN 2024

(i) Clotting of blood

(ii) Synthesis of thyroxine

18. Define the term-Diapedesis. MAIN 2024

19. Define the term hormone. MAIN 2024

20. The ____ glands produce hormones. COMP 2023

21. Adrenal medulla produces _____. MAIN 2023

22. The alpha cells of pancreas produce _____. MAIN 2022

23. Adrenal gland : Kidney : : Pituitary gland : _____ COMP 2023

24. Simple goitre is usually seen in people living in the hilly regions. Give two reasons for the statement. COMP 2021

25. What is Adrenal Virilism ? What causes this condition ? SEM II, 2021

26. Name the hormones whose deficiency causes : SEM II, 2021

(i) Diabetes mellitus

(ii) Diabetes insipidus

27. What is the cause of developing facial hair in some women? COMP 2023

Solution

1. Thus (c) is correct option.

2. The hormone that affects urination is: **(b) Vasopressin (ADH)**

3. Growth hormone (GH), secreted by the anterior pituitary, stimulates the growth of bones, muscles, and internal organs. **Hypersecretion of GH** leads to excessive growth: in children, it causes **gigantism**, and in adults, it results in **acromegaly**.

Hence, the correct answer is: **(d) Hypersecretion of Growth hormone.**

4. Growth hormone (GH) from the anterior pituitary regulates growth of bones, muscles, and internal organs. Hypersecretion of GH causes excessive growth: in children, it leads to gigantism, and in adults, it results in acromegaly.

Hence, the correct answer is: **(c) Hypersecretion of Growth hormone.**

5. Thyroid gland. It regulates basal metabolism.

6. Adreno cortico tropic hormone, Anterior pituitary gland secrete this.

7. Glycogen

8. Hormones or Adrenalin.

9. False

10. Islets of Langerhans are clusters of endocrine cells in the **pancreas** that regulate blood sugar by producing hormones like **insulin** and **glucagon**.

Hence, the correct answer is: **(d) Pancreas.**

11. When frightened, the body releases **adrenaline**, a hormone that prepares the body for **“fight or flight”** response by increasing heart rate, energy supply, and alertness.

Hence, the correct answer is: **(b) Adrenaline.**

12. Insulin

13. (i) Adrenaline secreted by the adrenal medulla brings about these changes (dry mouth, increased heartbeat).

(ii) **Endocrine glands** are glands that pour their secretion directly into the **blood**.

(iii) **Glucose is normally not found in urine** because insulin converts excess blood glucose into **glycogen**, which is stored in the **liver**.

(iv) **Hormonal reactions** are slower than nervous reactions because hormones travel through **blood**, while nerve impulses travel very fast through **nerves**.

(v) A diabetic person with high sugar in urine has **Glycosuria**.

14. ADH — Antidiuretic Hormone

Example of a diuretic substance: Alcohol (others include coconut water, caffeine).

15. Adrenaline is called 'The Emergency Hormone' because it is secreted under stress (anger, fear, excitement) and prepares the body to face emergency situations.

16. Tropic hormones stimulate other endocrine glands to secrete their specific hormones.

Example: TSH (Thyroid-Stimulating Hormone)

17. (i) Calcium

(ii) **Iodine**

18. Diapedesis is the process by which white blood cells squeeze through capillary walls to reach the site of infection.

19. Hormone is a chemical messenger produced by glands or specialized cells that circulates in body fluids to regulate physiological processes.

20. Endocrine (ductless)

21. Adrenaline and noradrenaline

22. Glucagon

23. Brain

24. Two reasons why simple goitre is seen in hilly regions:

(i) **Deficiency of iodine in the soil.**

(ii) **Iodine is required for the production of thyroxine.**

25. **Adrenal Virilism** is the development of male characteristics (beard, moustache, deep voice) in females.

Cause: Overgrowth of the adrenal cortex leading to excess secretion of the male hormone **androsterone**.

26. **Hormones whose deficiency causes:**

(i) Diabetes mellitus – Insulin

(ii) Diabetes insipidus – Antidiuretic hormone (ADH) or Vasopressin

27. The development of facial hair in some women is due to **overgrowth of the adrenal cortex**, which produces excess **male hormones**.